



REFERENCE MATERIAL



Chapter 4: The Calculus of Artificial
Neurons
Page 142 • Highlighted Section

-
-
-
-

MULTIPLE CHOICE

Which partial derivative represents the "error" signal propagated back to a hidden layer during training?

$\partial C / \partial w$



$\partial C / \partial a$



$\partial z / \partial x$



$\partial a / \partial z$



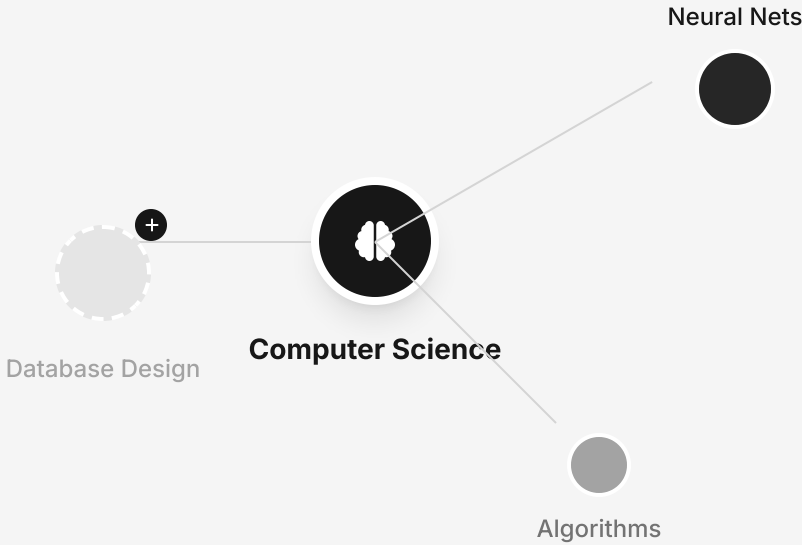
TAKE NOTE

EXPLAIN THIS

SAVE TO VAULT

Knowledge Graph

Visualization of your personal neural network



LEGEND

- Mastered
- Learning
- Identified Gap

Neural Networks

Sub-topic of Machine Learning

Mastery72%

KEY GAPS

- Vanishing Gradients
- LSTM Architecture

Study Topic

+

-



The Vault

 Filter sources by title, tag, or content...

All Types

Recently Added



PDF

Machine Learning Mastery

Added 2 days ago • 14.2 MB

AI

ML

SYNCED TO GRAPH



WEB

Understanding Vector Databases

Added 1 week ago • pinecone.io

Databases

PROCESSING AI...



VIDEO

React 19 Deep Dive

Added Oct 10 • 45:12

Frontend

READY



Add New Source

PDF, URL, or Text Note



Performance Analytics

Track your growth and retention over time

Week

Month

Year

TOTAL TIME

24.5h

↑ 12% vs last week

MASTERY SCORE

842

+ 45 new points

RETENTION RATE

92%

✓ Steady level

FOCUS SCORE

4.8/5

⚡ Top 5% users

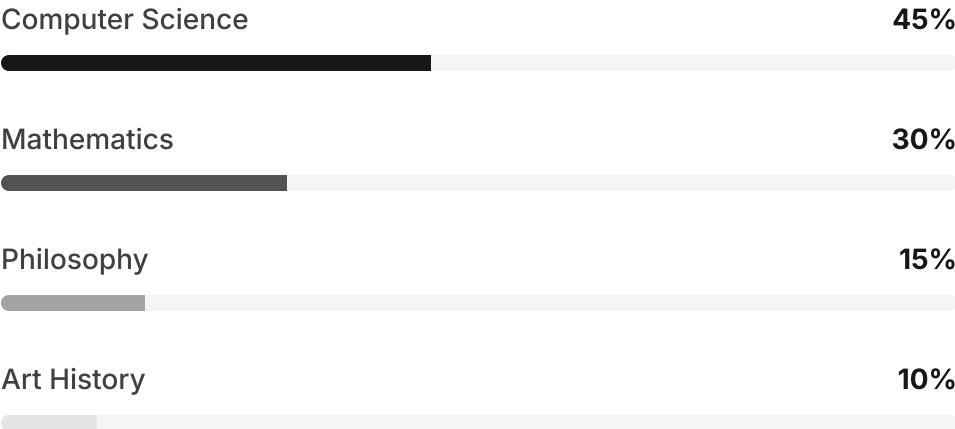
Knowledge Growth Curve

Actual Projected



KNOWLEDGE GROWTH GRAPH PLACEHOLDER

Topic Distribution



Activity Intensity



Settings



Profile



Notifications



Privacy & Security



Billing & Plan



Integrations



Sign Out

Public Profile

Manage how you appear on the platform.



Alex Carter

alex.carter@example.com

Change Photo

FIRST NAME

Alex

LAST NAME

Carter

BIO

Tell us about your learning goals...

Learning Preferences

Customize your AI-driven learning experience.

Adaptive Difficulty

AI adjusts session complexity based on your pace.



Spaced Repetition

Enable algorithmic scheduling for optimal retention.



🔍 | Search knowledge, sources, or concepts...

ESC

RECENT SEARCHES

Clear History

🕒 Neural Networks Architecture

🕒 Spaced Repetition Settings

SUGGESTIONS



Backpropagation

Concept in Knowledge Graph

72% Mastered



Deep Learning Book.pdf

PDF Document in Vault



Monthly Analytics

Application Screen

↵ to select

↑↓ to navigate

? for search syntax

Add New Content

×

Ingest information into your knowledge graph.



Upload Files

Drag and drop PDF, EPUB, or TXT files here. (Max 50MB)

Browse Files

WEB IMPORT

 Paste article URL (Medium, Wikipedia, etc.)

Import

DIRECT INPUT



Write a Note

Capture thoughts directly into the system.

➤

-  Daily Plan
-  Knowledge Graph
-  Vault
-  Analytics
-  Settings

Today's Plan

Wednesday, October 14, 2026

DAILY STREAK

12 days

75%

★ High Priority

Mastering Neural Networks: Backpropagation Explained

Next session: Quiz on gradient descent concepts and calculus fundamentals.

Start Session →

Reschedule



PREVIEW BLOCK

Upcoming Sessions

3 Tasks



Voice Practice: Italian Phrasing

Scheduled for 4:00 PM



React Performance Patterns

15 min read

Recently Finished

View All

- Intro to Vector Databases
- Microservices Architecture

88%

92%

TIME SPENT

1.2 hrs

KNOWLEDGE GAIN

+150 pts

CONFIDENCE

82%



Alex Carter
Pro Plan